Theory and Application of Transformers and Line Reactors

Engineering Technology

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Theory and Application of Transformers and Line Reactors (A18281)

Short Title:	Transformer & Reactor Theory		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Engineering		
Author	NEMURPHY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The aim of this module is to provide the student with a professional level of understanding, and the ability to apply associative design and calculation methods to the areas of MV and HV power transformers and line reactors on transmission and distribution power networks. In addition, the student will gain proficiency in the areas of safety procedures, and maintenance requirements of transformers and reactors.

Programmes

BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	4 / 7 / M
BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 7 / M
BEng (Hons) (`WD_ETRIC_B`)	4 / 7 / M

Indicative Content

- Classification of transformers and line reactors
- Phase shift transformer theory & applications of tertiary windings
- Shunt/series, capacitive, and inductive reactor rating & design.
- Onload & offload transformer/reactor tap-changing methods and applications
- Core assembly, winding practices, insulation types, and cooling methods.
- Current inrush, over-voltage, and short-circuit protection.
- CELNEC, IEEE/ANSI, and IEC Standardisation of transformers and reactors.
- Transformer/reactor loss capitalisation, and optimum design criteria

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate and understand the design and operation of power transformers and reactors on HV and MV sectors of a national electrical grid.
2. Configure the tertiary winding of a transformer for phase shift, and load balancing applications.
3. Determine appropriate tap changer designs for both on-load and off-load configurations.
4. Demonstrate proficiency in the methods of core assembly, winding practice, insulation type, and cooling methods of inductor hardware.
5. Apply theoretical background to the detection and suppression of current inrush and other parameters that exceed inductor rating.
6. Generate costing for optimum design criteria and capitalisation of inductor loss.
7. Provide confirmation of compliance with CELNEC, IEEE/ANSI, and IEC Standardisation in terms of operation and safety.

Learning and Teaching Methods

- The module delivery method will consist of lectures, laboratories, mini-project, and formative assessments.
- The emphasis on module delivery is to engage the student in the learning of the module content through the integration of course materials with case studies in laboratories and the application within a mini-project.
- The formative assignments will provide an opportunity for the student to engage with the material as it is being applied to maximise the learning opportunities of the student.
- Self-directed learning, moodle, and recommended software applications optimise the delivery of module material in specific areas where additional processes are required.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Project	10%	2,6,7
Practical	10%	6,7
Case Studies	10%	3,6,7

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Harlow, J.H. _Electrical Power Transformer Engineering_. 3rd ed. US: CRC Press, May 2012.

Supplementary Material

- Das, J.C. _Transients in Electrical Systems: Analysis, Recognition, and Mitigation_. 1st. US: McGraw-Hill, 2010.
- Del Vecchio, R.M., B. Poulin, P. Feghali, M.S. Dilipkumar and A. Rajendra. _Transformer Design Principles: With Applications to Core-Form Power Transformers_. 2nd ed. US: CRC Press, May 2010.